

MAT1503

October/November 2019

Linear Algebra

Duration 2 Hours

100 Marks

EXAMINERS :

FIRST

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Closed book examination

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This paper consists of 4 pages

ANSWER ALL QUESTIONS.

ALL CALCULATIONS MUST BE SHOWN.

QUESTION 1

(a) (i) Let $A = \begin{bmatrix} 1 & \frac{1}{3} \\ c & d \end{bmatrix}$ Find numbers c and d such that $A^2 = 0$. (3)

(ii) Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$ Find A^{32} (2)

(b) Let D and E be $n \times n$ matrices Prove that DE is invertible if and only if D and E are both invertible. (4)

(c) Let $A = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$

(i) Find A^{-1} by using row operations. (4)

(ii) Use part (c)(i) to express A as a product of elementary matrices (4)

(d) Let A be an $n \times n$ matrix and let A^T denote the transpose of A
Show that

(i) $A + A^T$ is symmetric (2)

(ii) $A - A^T$ is skew-symmetric (2)

(iii) $(A^T)^{-1} = (A^{-1})^T$ if A is invertible (4)

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QUESTION 2

(a) Find the values of k for which the system of equations

$$x + ky = 1$$

$$kx + y = 1$$

has

(i) no solution

(ii) exactly one solution

(iii) infinitely many solutions (7)

(b) Solve the following homogeneous system of equations

$$\begin{aligned} x + 2y + 3z &= 0 \\ 2x + 5y + 2z &= 0 \\ x + 8z &= 0 \end{aligned} \quad (5)$$

(c) Use Cramer's Rule to solve for x in the following system of linear equations

$$\begin{aligned} -x + 3y - 2z &= 5 \\ 4x - y - 3z &= -8 \\ 2x + 2y - 5z &= 7 \end{aligned} \quad (8)$$

(d) Using the determinant, determine the values of λ for which the following matrix A has an inverse, A^{-1}

$$A = \begin{bmatrix} \lambda - 1 & 0 & -(\lambda - 1) \\ 0 & 1 & 0 \\ 2 & 1 & \lambda + 1 \end{bmatrix} \quad (5)$$

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QUESTION 3

Consider the vectors

$$\underline{u} = (-2, 1) \quad \text{and} \quad \underline{v} = (1, -2)$$

(a) Determine $\cos \theta$, where θ is the angle between $\underline{u}$ and $\underline{v}$ (4)

(b) Determine the area of the parallelogram determined by $\underline{u}$ and $\underline{v}$ (3)

(c) (i) Calculate $\underline{u} \times \underline{v}$ (4)

(ii) What is the angle between $\underline{u}$ and $\underline{v}$? (2)

(d) Find the plane that passes through the point $(2, 4, -3)$ and is parallel to the plane $-2x + 4y - 5z + 6 = 0$. (4)

(e) Find the line that passes through the point $(2, 5, 3)$ and is perpendicular to the plane $2x - 3y + 4z + 7 = 0$. (4)

- (f) Find an equation of the plane passing through the point $(-2, 3, 4)$ and is perpendicular to the line passing through the points $(4, -2, 5)$ and $(0, 2, 4)$ (4)

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QUESTION 4

- (a) State De Moivre's Theorem (2)

- (b) Express $\cos 3\theta$ and $\sin 3\theta$ in terms of sines and cosines of multiples of θ . (9)

- (c) Let w be a complex number, z a 4th root of w

- (i) Show that $z(k) = \rho^{\frac{1}{4}} \left[\cos \left(\frac{\phi + 2k\pi}{4} \right) + i \sin \left(\frac{\phi + 2k\pi}{4} \right) \right]$, $k = 0, 1, 2, 3$ is a formula for the 4th roots of w , where ϕ is the argument of w and ρ is its modulus

Show all your working. (8)

- ((ii) Hence determine the 4th roots of 16. (6)

[25]

TOTAL: [100]